Data availability

Chemical
pH (1:5)
E.C. (mS/cm)
Macronutrients (N,Ca, P...)
Micronutrients (Fe,Mb,Cu...)
Carbonates (% CO_3)
CEC (meq/100g)

Biological
Organic Matter (%)
C/N Ratio
Organic carbon (%)

Physical
Clays (%)
Silts (%)
Sands (%)
Texture

Risks

Signal-to-noise in SOC is addressed through multi-year sampling windows, consistent laboratories/methods and composite sampling over adequate area. Yield or weed surprises during transition are mitigated with locally adapted cover-crop mixes and carefully timed termination that protects the following cash crop, backed by accessible advisory support. Administrative load is kept lean by aligning MRV calendars with farm operations, using standard templates and accepting digital submissions. Permanence and liability are managed through rolling contracts with renewal options, programme-level buffer accounts and clauses that fairly account for uncontrollable weather shocks.



Preliminary results

At scale, the CiRAA scheme is expected to improve topsoil structure, enhance infiltration and water storage and deliver stable or rising SOC in the upper profile—while preserving commercially viable rotations and quality. The same evidence base enables cleaner contracts: realistic targets, clear baselines and proportionate checks so farmers know what is expected and agencies or buyers can defend payments.

Economic perspective

Public **action-based payments** cover additional inputs and a portion of income risk during transition. The result-based top-up then pays for maintained or improved performance relative to the baseline, with payment curves that recognise plateaus (rewarding the preservation of good status) and share uncertainty through modest buffers for inter-annual variability. Transitional grants can target first-time investments (e.g., seeders/rollers, cover-crop seed), while cooperatives lower unit costs via pooled procurement and shared machinery.

Policy and governance

The governance framework is embedded within the CAP architecture but refined to reduce administrative complexity and ensure accountability. The regional CAP paying agency defines eligibility rules, issues action-based contracts (e.g. organic management, reduced tillage, cover crops), and processes farmer claims, while a PES operator manages the result-based top-ups, establishes performance thresholds, and oversees verification. Independent research partners act as neutral knowledge brokers, providing technical protocols, QA/QC, and ensuring that indicators remain agronomically sound and proportionate. Farmers and cooperatives carry out the practices, maintain concise field logs, and co-design operational calendars for sowing, termination, and harvest, while third-party certifiers conduct light-touch audits that integrate seamlessly with paying-agency workflows. This clear division of responsibilities not only reduces duplication and transaction costs but also strengthens trust between stakeholders, particularly in the critical early phase of baseline establishment.

CiRAA LTEs on conventional and organic agriculture- Payment for Ecosystem Services

The CiRAA long-term experiments (LTEs) at the University of Pisa constitute one of Italy's most valuable living laboratories for soil-health management. Running continuously since the mid-1980s across substantial field scales, the LTEs compare conservation agriculture (reduced or no tillage, cover crops) with conventional mouldboard ploughing, and juxtapose organic and conventional whole-farm systems under realistic rotations. The case study's ambition is to translate this rare continuity of biophysical and economic evidence into a Payments for Ecosystem Services (PES) architecture .

novasoil

The NOVASOIL project aims to show the benefits for society and the environment of investing in soil health.

The main output of the project will be a tool to analyse the suitability of different business cases that promote soil health.

This tool will be based on best practice examples in Europe and other countries, as well as on the needs and demands of society.

A model for multifunctional and sustainable development of marginal areas

Summarize



The case study derives from an integrated supply chain project funded in the framework of EU's Rural Development Policy, which involved research centres, private companies, farmers, and public administrations, to drive force of rural sustainable development through new cropping systems and management innovations. The internal and hilly areas of Tuscany Region are marginal areas characterized by conventional extensive agriculture, mainly rely on cereal crops, and little agricultural value. Soil erosion, nutrient leaching, reduction of C in soils, certainly represent the main critical issues in these areas.

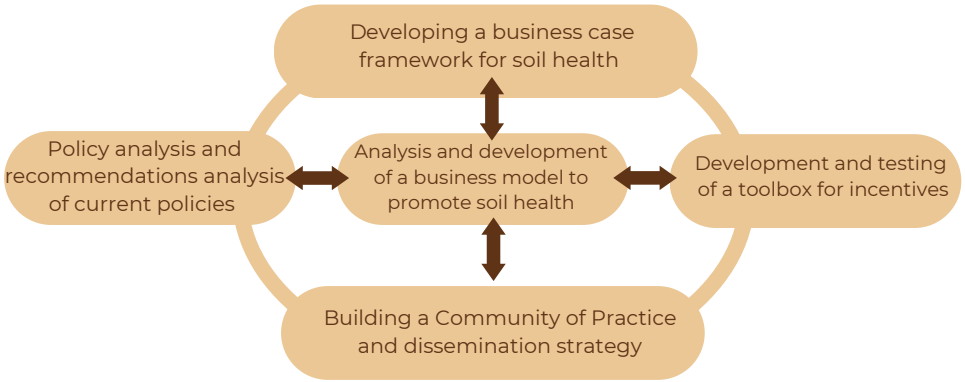


Figure 1. Structure of the NOVASOIL project

Objectives

In the NOVASOIL Project, we will focus on learning from experiences in the project by exploring the effectiveness of the actions made in terms of soil fertility and health improvement, productivity rising, resilience and landscape valorisation. We will investigate the possibility to develop a business models aiming at adding value to the agronomic management aimed at soil health conservation with innovative crop products through certification and production standards.

Case studies properties

Potential inhibitors

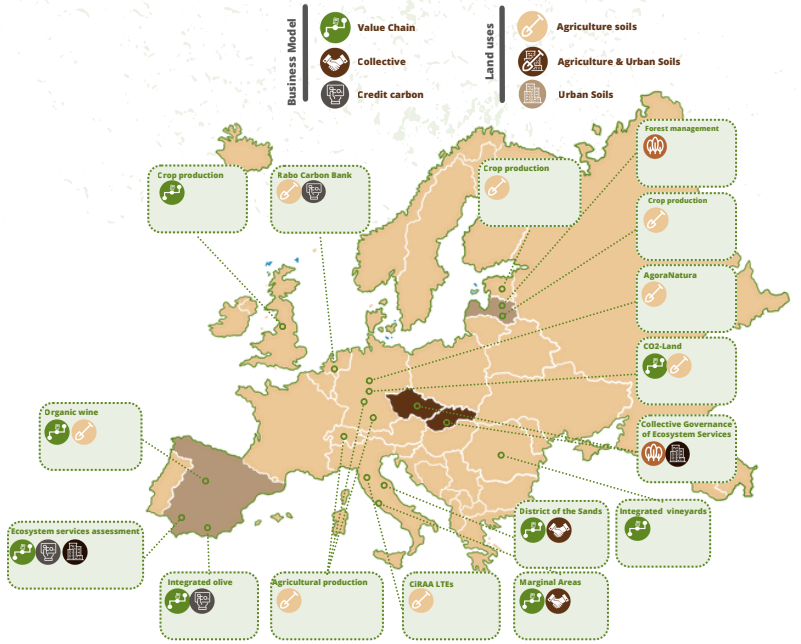
Competition with products from other countries with a lower cost,
Market changes
Extreme weather conditions (extreme rainfall, floods, and drought)

Potential enablers

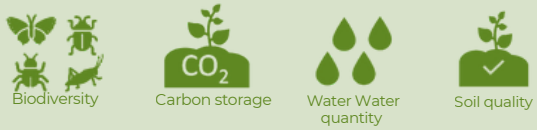
Local, organic certified organic and fully traceable final products
Social awareness interested in high quality and healthy product
Existing farmer network
Higher revenue for farmers compared to previous cropping system

Potential stakeholders

Tuscany Region - Regional administration | Private companies | Farmers |Farmer association | Accademia dei Georgofili



Ecosystem services



Specific objectives

SO1 Multi-dimensional portfolio development of business cases and good examples for investing in soil health in Europe and beyond.

SO2 Analysis of current models around the world for creating new incentives and new revenues from healthy soils.

SO3 Apply a co-development approach to modelling that promotes soil health.

SO4 Develop a prototype incentive toolbox and test network.

SO5 Analysis of current policies and technologies related to soil health and income provision for farmers.

SO6 Developing a community of practice around investors and key soil health actors.

SO7 Increase interest in soil health through a digital marketing strategy plan

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INNOVATIVE BUSINESS MODELS FOR SOIL HEALTH

https://novasoil-project.eu

